

Spectroscopy as a Probe of Spacetime Foam

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Quantum gravity predicts that spacetime fluctuates stochastically at the Planck scale — a “foam” that imprints a tiny achromatic arrival-time jitter on light from distant sources.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the proper distance to the source, α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

We develop a framework based on the first-order coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. The first-order coherence – the field–field autocorrelation, equivalently the fringe visibility at delay τ – inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows, extending the binary Lieu–Hillman erasure criterion [7] ($\sigma_\phi \gtrsim 1$) to a continuous curve and recovering the Strehl and fringe-visibility tests as limiting cases of the same statistic.

The remainder of the paper is organized as follows.

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II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for virtually all known astrophysical sources) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

In the geometric-optics limit the path-length shift $\Delta \ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity is a key discriminant between foam and conventional propagation effects.

B. Spatial and Temporal Fluctuations of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — quantum foam models posit that each Planck-scale cell along the line of sight fluctuates in time. Two photons traversing the same cell at coordinate times separated by τ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\text{foam}}$ but decorrelated for $\tau \gg \tau_{\text{foam}}$, where τ_{foam} is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_P$ cells along the line of sight. The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. If τ_{foam} is of order the Planck time, $\ell_P/c \approx 5 \times 10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\text{foam}} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all τ of interest and every C_α -dependent temporal observable vanishes: the finite-lag Siegert discrepancy $\mathcal{S}(\tau)$ is suppressed by $[1 - C_\alpha(\tau)] \rightarrow 0$ (Eq. (??)), and the single-ray $g_{\text{obs}}^{(1)}$ recovers its intrinsic value because the foam suppression factor $\mathcal{F}_{\text{foam}} = e^{-\sigma_\phi^2[1-C_\alpha]} \rightarrow 1$ (Eq. (24)). In this quasi-static limit the HBT framework loses sensitivity altogether and the foam manifests only through the imaging-channel $g^{(1)}$ suppression on time-averaged exposures. The full spatiotemporal covariance between two rays separated by transverse distance b takes the factorized form

$$\langle \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}) \Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}') \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \quad (4)$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (so $b = 0$):

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (5)$$

This product form — separating the total variance from the normalized spatiotemporal shape — is a structural assumption, not a consequence of Eq. (1), that defines a tractable class of foam models within which the analytic framework is exact. The parameters A_α , $C_\alpha(\tau; D)$, τ_{foam} , $r_\perp$, and Φ_{eff} parameterize distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and each governs a distinct aspect of the signal's detectability. The factorization is consistent with phenomenologies in which the longitudinal path-length variance, the temporal decorrelation, and the transverse coherence arise from statistically independent mechanisms — e.g. the simple cellular pictures of Refs. [4–6] where each Planck-scale cell contributes an independent random kick.

The factorization is *not* generic: the pixellon model of Lee, Zurek & Chen [10] has long-range angular correlations whose spatiotemporal covariance does not reduce cleanly to $\sigma_\ell^2(D) C_\alpha(\tau) \Phi(b/r_\perp)$; the framework here applies to that model only as a heuristic approximation in the regime where transverse correlations have decoupled.

Observationally, factorization predicts that the extended-source Siegert discrepancy is suppressed relative to its point-source value by the angular-size factor $\Phi_{\text{eff}}(r_{\perp})$: $\mathcal{S}_{\text{ext}}(\tau) \approx \Phi_{\text{eff}} \mathcal{S}_{\text{pt}}(\tau)$ in the weak-foam regime (Section ??), independent of both τ and D . A dependence of this ratio on lag or source redshift would falsify the factorization ansatz and indicate that a non-factorizing model (e.g. pixellon) is required. The single observable developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ — descends from Eq. (1), with Sections III–?? probing its longitudinal projection (single-ray temporal jitter) and Section ?? its transverse projection (cross-ray Φ_{eff} suppression).

C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t + \tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time t :

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}'). \quad (6)$$

For fixed D , stationarity implies the statistics depend only on τ and the transverse separation $b(\boldsymbol{\theta}, \boldsymbol{\theta}') = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}) - \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}')]^2 \rangle \\ &= \frac{1}{c^2} \left[\sigma_{\ell}^2 + \sigma_{\ell}^2 - 2\sigma_{\ell}^2 C_{\alpha}(\tau) \Phi\left(\frac{b}{r_{\perp}}\right) \right] \\ &= \frac{2\sigma_{\ell}^2(D)}{c^2} \left[1 - C_{\alpha}(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_{\perp}}\right) \right]. \end{aligned} \quad (7)$$

For $b \gg r_{\perp}$, $\Phi \rightarrow 0$ and the variance becomes $2\sigma_{\ell}^2/c^2$ at all lags, including $\tau = 0$ — a new effect absent in the point-source case, and one that is independent of the temporal fluctuation structure of the foam (i.e. independent of C_{α}). In the dynamical limit $C_{\alpha} \rightarrow 0$ this result holds for all separations b , independently of Φ .

By the central limit theorem (CLT), $\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \sim \mathcal{N}(0, \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}')), \quad (8)$$

where σ_{Δ}^2 is taken from Eq. (7).

The Gaussian approximation follows from the CLT: for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_{\text{P}} \gtrsim 10^{46}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For the cross-ray case the additional condition $b \gg r_{\perp}$ is assumed, and is satisfied in the fully-resolved limit $\Phi_{\text{eff}} \rightarrow 0$ adopted in Section ??.

III. FIRST-ORDER COHERENCE

A. Point-Source Differential Jitter

In this section we restrict to a point source ($b = 0$), for which the cross-ray differential jitter of Eq. (6) reduces to the single-ray differential jitter

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t), \quad (9)$$

with variance obtained from Eq. (7) at $\Phi(0) = 1$:

$$\sigma_{\Delta}^2(\tau; D) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - C_{\alpha}(\tau; D)]. \quad (10)$$

Although this section restricts to a point source for notational compactness, this single-ray variance applies equally to extended sources at finite lag $\tau \neq 0$: under the factorized covariance Eq. (4), the temporal factor $C_{\alpha}(\tau; D)$ is identical for all ray pairs, so cross-ray transverse structure enters only at $\tau = 0$ through Φ_{eff} (Section ??).

B. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (11)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (12)$$

Since δt is independent of E_{src} , averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (13)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

C. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}, \quad (14)$$

where τ_L is the coherence time. Evaluating Eq. (13) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = e^{\sigma_{\Delta}^2/2\tau_L^2} \cdot \frac{1}{2} \left[e^{-\tau/\tau_L} \text{erfc}\left(\frac{\sigma_{\Delta}/\tau_L - \tau/\sigma_{\Delta}}{\sqrt{2}}\right) + e^{\tau/\tau_L} \text{erfc}\left(\frac{\sigma_{\Delta}/\tau_L + \tau/\sigma_{\Delta}}{\sqrt{2}}\right) \right]. \quad (15)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \tau_L$, the Gaussian average of $e^{-|\tau - \Delta|/\tau_L}$ over $\Delta \sim \mathcal{N}(0, \sigma_{\Delta}^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2\tau_L^2} + \mathcal{O}(\sigma_{\Delta}^4) \right]. \quad (16)$$

Here $\sigma_{\Delta}(\tau)$ is τ -dependent through $C_{\alpha}(\tau)$, and the expansion is in $\sigma_{\Delta}(\tau)/\tau_L$ at fixed τ . The correction is positive because $e^{-|\tau|/\tau_L}$ is convex and Jensen's inequality applies. The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved exactly.

For a Voigt profile $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width τ_L (from optically-thick electron scattering or BLR turbulent broadening) convolved with a Gaussian of width τ_G (from bulk Doppler motions) — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde{\sigma}_1^2 = \frac{\sigma_{\Delta}^2 \tau_G^2}{\sigma_{\Delta}^2 + \tau_G^2}, \quad \mu_1(\tau) = \frac{\tau_G^2 \tau}{\sigma_{\Delta}^2 + \tau_G^2}, \quad (17)$$

giving, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_{\Delta}} e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)} e^{\tilde{\sigma}_1^2/2\tau_L^2} \frac{1}{2} \left[e^{-\mu_1/\tau_L} \text{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L} \text{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) \right]. \quad (18)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \min(\tau_L, \tau_G)$, expanding Eq. (13) to second order in Δ and averaging over the Gaussian jitter gives the Voigt analogue of Eq. (16):

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2} \left(\left(\frac{1}{\tau_L} + \frac{\tau}{\tau_G^2} \right)^2 - \frac{1}{\tau_G^2} \right) + \mathcal{O}(\sigma_{\Delta}^4) \right], \quad (19)$$

which reduces to Eq. (16) in the purely Lorentzian limit $\tau_G \rightarrow \infty$. Unlike the Lorentzian case, the correction factor can change sign: for $\tau < \tau_G \sqrt{1 - \tau_G/\tau_L}$ the second-order term is negative, so foam jitter *suppresses* short-lag coherence. Note that both closed forms in the weak-jitter limit share the general product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) F_{\text{foam}}(\tau)$.

D. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, $E_{\text{src}}(t) = A(t) e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c \Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t - \delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\text{obs}}(t) \approx A(t) e^{-i\omega_0 t} e^{i\omega_0 \delta t(t)}. \quad (20)$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is ≈ 0.28 , so the exact convolution Eq. (15) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Section ??), so Eq. (22) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0 ; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \quad (21)$$

This holds when $\tau \gg \tau_{\text{foam}}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\text{foam}} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (10):

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \exp(-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]), \quad (22)$$

where the rms accumulated phase follows from the Ng–Perlman path-length model (1) via the phase conversion $\sigma_\phi = (2\pi/\lambda_{\text{obs}}) \sigma_\ell(D_C)$, with D_C the line-of-sight comoving distance [4]:

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}. \quad (23)$$

We write the foam suppression factor — the quantity appearing in the product structure of Section III C — as

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \quad (24)$$

so $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \mathcal{F}_{\text{foam}}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2 [1 - C_\alpha(\infty; D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ is compatible with the weak-jitter expansion of Eq. (16) — the former controls envelope displacement, the latter the carrier-phase suppression — and when neither applies the full convolution Eq. (13) must be evaluated.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (13) with closed forms Eqs. (15)–(18) — carries direct spectroscopic content: for any wide-sense stationary (WSS) field, the normalized power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\{g_{\text{obs}}^{(1)}(\tau)\}(\nu). \quad (25)$$

The theorem applies here for two reasons. First, astronomical emission lines have finite linewidth: their spectra fall to zero at both low and high frequencies, so $\Gamma_{\text{src}}(\tau) \rightarrow 0$ and $\Gamma_{\text{src}} \in L^1$; since the foam factor is bounded, $\Gamma_{\text{obs}} \in L^1$ as well and the Fourier transform is well defined. Second, $E_{\text{obs}}(t)$ is wide-sense stationary: the foam-induced jitter $\delta t(t)$ is a stationary process (Eq. (5), Section II) that is statistically independent of the source, so $\Gamma_{\text{obs}}(\tau)$ — given in general by Eq. (13) — depends only on the lag τ , provided $T_{\text{int}} \gg \tau_{\text{foam}}$ so that the foam statistics are ergodically sampled. In the further limit where the foam decorrelates on long time scales ($C_\alpha(\tau) \rightarrow 0$ as $|\tau| \rightarrow \infty$, the dynamical regime), the foam factor acquires a nonzero asymptote $\mathcal{F}_{\text{foam}}(\tau) \rightarrow e^{-\sigma_\phi^2}$, generating a distributional $\delta(\nu)$ contribution that encodes the fraction $e^{-\sigma_\phi^2}$ of source flux propagating phase-coherently.

When the lag-domain product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \mathcal{F}_{\text{foam}}(\tau)$ holds—the weak-jitter and narrow-band regimes—the convolution theorem applied to Eq. (25) yields

$$S_{\text{obs}}(\nu) = S_{\text{src}}(\nu) * \mathcal{K}_{\text{foam}}(\nu). \quad (26)$$

i.e. the intrinsic spectrum is convolved by a kernel $\mathcal{K}_{\text{foam}}(\nu) \equiv \mathcal{F}\{\mathcal{F}_{\text{foam}}\}(\nu)$ that depends on both the foam (through σ_ϕ^2 , C_α , τ_{foam}) and the source line.

B. Connection to Fringe Visibility and Strehl Ratio

The first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (24). The fringe visibility and Strehl ratio are the *spatial* counterparts: they probe phase variations between two points in the telescope aperture plane separated by a baseline B , with the suppression governed by an aperture-plane coherence function $C_\alpha^{\text{sp}}(B; D)$.

The structural equivalence between the temporal and spatial cases is not merely formal: in both, the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable, with C_α (respectively C_α^{sp}) setting how that variable's variance grows with separation. Temporally, the relevant random variable is the differential jitter $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$ between two arrival times along a single ray, with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (10)). Spatially, the analogous random variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (27)$$

Applying the Gaussian characteristic-function identity of Eq. (21) in each domain yields suppression factors of identical form—one expressed in lag, the other in baseline:

$$\mathcal{F}_{\text{foam}}^{\text{sp}}(B) \equiv \exp\{-\tfrac{1}{2}\sigma_{\Delta\phi}^2(B)\} = \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (28)$$

mirroring $\mathcal{F}_{\text{foam}}(\tau)$ of Eq. (24) under $C_\alpha(\tau; D) \leftrightarrow C_\alpha^{\text{sp}}(B; D)$. The shared subscript α further reflects that the same cumulative growth law of Eq. (1) underlies both decorrelations. As with its temporal counterpart, $\mathcal{F}_{\text{foam}}^{\text{sp}}$ runs from unity at zero baseline ($C_\alpha^{\text{sp}}(0; D) = 1$, the two points sample identically phased wavefronts) to maximum suppression $e^{-\sigma_\phi^2}$ in the resolved limit where $C_\alpha^{\text{sp}} \rightarrow 0$ (the two points sample independent foam realizations).

The two-element interferometric fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \mathcal{F}_{\text{foam}}^{\text{sp}}(B) = V_{\text{src}}(B) \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (29)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}|$ is evaluated at angular scale λ/B . The Strehl ratio of a single aperture of diameter D_{tel} is the same statistic with B replaced by the internal aperture baseline: when $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$ the aperture-plane phases are independent draws from the single-ray distribution and the Maréchal formula gives [4]

$$S \simeq e^{-\sigma_\phi^2}. \quad (30)$$

Strehl ratio and fringe visibility therefore measure the *same* underlying phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the resolved limit, differing only in whether what part of the point-spread-function is being considered. This unification places the published imaging constraints [4, 7–9] on a common footing within the present framework.

Equation (29) also extends the binary Lieu–Hillman [7] erasure criterion to a continuous suppression curve. Their condition $\Delta\phi \gtrsim 2\pi$, with $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$, is precisely $\sigma_\phi \gtrsim 1$ —the saturation limit $\mathcal{F}_{\text{foam}}^{\text{sp}} \rightarrow e^{-\sigma_\phi^2} \rightarrow 0$ of Eq. (28). The shape of the continuous curve, observed as a function of baseline, encodes the aperture-plane coherence profile $C_\alpha^{\text{sp}}(B; D)$, in direct analogy to how $g_{\text{obs}}^{(1)}(\tau)$ encodes the temporal decorrelation $C_\alpha(\tau; D)$. Two observational handles follow from this correspondence: (i) the chromatic scaling $\sigma_\phi \propto 1/\lambda$ implies that constraints tighten rapidly with decreasing wavelength—a fact exploited by Ng & Perlman [4] (their Table 1) in moving from optical through X-ray and γ -ray observations, and mirrored in Section V B, where the same saturated-limit suppression $e^{-\sigma_\phi^2}$ that defines the Strehl ratio (Eq. (30)) is reinterpreted as the coherent fraction of a spectral line, yielding a spectroscopic counterpart of the Strehl bound at X-ray wavelengths; (ii) baseline-resolved visibility measurements of a single target can in principle probe the aperture-plane coherence scale, complementary to the temporal decorrelation scale accessed through $g_{\text{obs}}^{(1)}(\tau)$.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are blurred, in direct analogy to the blurring of the point-spread function in imaging. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- y distortions. In this section we consider these two possibilities.

A. GRB 221009A

GRB 221009A (“the BOAT,” Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\text{iso}} \sim 10^{55}$ erg at redshift $z = 0.151$, a once-in- $\sim 10^4$ -year event [11]. The burst was detected from radio wavelengths through ≈ 18 TeV (LHAASO). Two independent spectral analyses have reported a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [12, 13], at observed energies in the range 10–37 MeV. We apply the spectral-Strehl bound Eq. (34) to each detection. Adopting flat Λ CDM ($\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the comoving distance is $D_C(z = 0.151) \approx 0.63 \text{ Gpc} \approx 1.93 \times 10^{25} \text{ m}$ and $\ln(D_C/\ell_P) \approx 138.3$.

Ravasio et al. (2024). Ravasio et al. [12] report a 13σ detection of a narrow emission feature in Fermi/GBM data from the intervals $T+280$ – 320 s, outside the bad-time interval affected by detector saturation. The line centroid evolves from ≈ 14 MeV at $T+280$ s to ≈ 6 MeV at $T+320$ s; the most energetic well-constrained single measurement is $E_{\text{line}} = 12.56 \pm 0.30$ MeV in the interval $T+280$ – 300 s (improvement $\Delta\text{AIC} = 49$ over the continuum-only model). The probe wavelength is $\lambda_{\text{obs}} = hc/E_{\text{line}} \approx 9.9 \times 10^{-14} \text{ m}$. The Gaussian width measured in this interval is $\sigma_{\text{gauss}} = 1.31^{+0.30}_{-0.31} \text{ MeV}$, giving a relative width $\sigma/E \approx 10\%$. The narrow-band condition underlying Eq. (26) is satisfied: at the bound $\alpha \approx 0.65$ the rms path-length is $\sigma_\ell \approx 1.6 \times 10^{-14} \text{ m}$, while the coherence length $c\tau_c \approx \hbar c/\sigma_E \approx 2.1 \times 10^{-13} \text{ m}$, giving $\sigma_\ell/(c\tau_c) \approx 0.07$; the envelope approximation is accurate to $\lesssim 1\%$. Inserting $D_C \approx 1.93 \times 10^{25} \text{ m}$, $\lambda_{\text{obs}} \approx 9.9 \times 10^{-14} \text{ m}$, and $\ell_P = 1.6 \times 10^{-35} \text{ m}$ into Eq. (34): $\ln(2\pi D_C/\lambda_{\text{obs}}) \approx 90.0$ and $\ln(D_C/\ell_P) \approx 138.3$, giving

$$\alpha \gtrsim \frac{90.0}{138.3} \approx 0.65. \quad (31)$$

Zhang et al. (2024). Zhang et al. [13] perform a joint analysis of GECAM-C and Fermi/GBM data covering the full prompt emission, including the bright phase inaccessible to GBM alone. They identify a spectral excess in the G1 time interval ($T+246$ – 256 s) at ≈ 37 MeV, and interpret the entire feature as a line whose centroid evolves from ≈ 37 MeV down to ≈ 6 MeV across the burst. The 37 MeV detection lies at or beyond the stated upper boundary of the GBM BGO detector range (4–35 MeV), so the centroid and width of this earliest interval carry larger systematic uncertainty than the Ravasio et al. detections. Taking $E_{\text{line}} \approx 37$ MeV at face value gives $\lambda_{\text{obs}} \approx 3.4 \times 10^{-14} \text{ m}$, $\ln(2\pi D_C/\lambda_{\text{obs}}) \approx 91.1$, and

$$\alpha \gtrsim \frac{91.1}{138.3} \approx 0.66. \quad (32)$$

This bound should be regarded as tentative pending confirmation of the 37 MeV excess as a narrow spectral line within the instrumental sensitivity range.

Both bounds approach but do not exclude the holographic value $\alpha = 2/3 \approx 0.667$: at that exponent $\sigma_\phi^2(\alpha = 2/3) \approx 0.01$ for the Ravasio line, corresponding to a coherent fraction $e^{-0.01} \approx 99\%$ — the line would be essentially unaffected. Formally ruling out $\alpha = 2/3$ at $z = 0.151$ would require a confirmed spectral line at $E_{\text{obs}} \gtrsim 115$ MeV, approximately three times higher than the most energetic detection reported to date.

B. BzK4892

The two-component line-profile decomposition of Eq. (26) admits a direct application as a spectroscopic analogue of the Strehl-ratio bound [4]: just as the observation of a diffraction-limited point-spread function at wavelength λ and distance D requires the Strehl ratio $\mathcal{S} \simeq e^{-\sigma_\phi^2}$ (Eq. (30)) to remain of order unity, i.e. $\sigma_\phi^2 \lesssim \mathcal{O}(1)$ (§3 and Eq. (4) of Ref. [4]), the observation of sharp spectral features requires the same condition: in the dynamical foam limit the lag-asymptote $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$ of Eq. (24) plays the role of a *spectral Strehl* — the fraction of source flux delivered into the coherent line core $e^{-\sigma_\phi^2} S_{\text{src}}(\nu)$ of Eq. (26) — and for $\sigma_\phi^2 \gtrsim 1$ this coherent core is suppressed below the incoherent pedestal so that crisp line profiles cease to be detectable. The two arguments are structurally identical: each is a single-number, saturated-limit test of the coherent fraction $e^{-\sigma_\phi^2}$, differing only in whether that fraction is read off the on-axis PSF intensity or off the spectral line core. They share the same threshold criterion and the same parametric dependence on $\{D, \lambda, \alpha\}$.

BzK 4892 is a massive ($M_\star \approx 5 \times 10^{10} M_\odot$), infrared-luminous, dust-obscured star-forming *BzK* galaxy in the CDF-S at spectroscopic redshift $z_s = 2.578$, hosting a heavily obscured, intrinsically quasar-luminous ($L_{2-10 \text{ keV}} \gtrsim 10^{44.5} \text{ erg s}^{-1}$) AGN behind a Compton-thick absorber with $N_{\text{H}} \sim 10^{24.5-25} \text{ cm}^{-2}$ [14]. Because the direct hard X-ray continuum is suppressed by photoelectric absorption and Compton down-scattering, the X-ray spectrum is reflection-dominated and the 6.4 keV Iron K α fluorescence line— produced by the obscurer reprocessing the buried

AGN continuum—emerges with exceptionally large rest-frame equivalent width, $EW \simeq 2.3 \text{ keV}$, detected at $\gtrsim 4\sigma$ in the *Chandra* 4 Ms CDF-S exposure [14]. The redshift is independently confirmed by multiple narrow rest-UV emission lines (Ly α , C IV, He II, C III) detected in VLT/FORS spectroscopy [15, 16]. Adopting the same flat Λ CDM cosmology, the line-of-sight comoving distance is $D_C(z_s = 2.578) \approx 6.1 \text{ Gpc} \approx 1.88 \times 10^{26} \text{ m}$. The spectroscopically detected Iron K α 6.4 keV emission line, redshifted to an observed energy of $6.4/(1+z_s) \approx 1.79 \text{ keV}$ ($\lambda_{\text{obs}} \approx 6.93 \times 10^{-10} \text{ m}$), provides the probe wavelength for this estimate; the short rest-frame wavelength ($\lambda_{\text{rest}} \approx 1.94 \text{ \AA}$) makes the foam variance $\sigma_\phi^2 \propto \lambda_{\text{obs}}^{-2}$ much larger than for any optical line at the same distance, yielding a strong constraining spectroscopic bound.

The threshold condition $\sigma_\phi^2 \lesssim 1$ for the coherent line core of Eq. (26) to remain detectable above the foam-broadened pedestal — the spectral-Strehl criterion introduced above, mirroring the imaging Strehl bound of Eq. (30) — depends on the intrinsic source profile $S_{\text{src}}(\nu)$ only through its characteristic width $\Delta\nu_{\text{src}}$, and not through its detailed functional form. For the fluorescent Iron K α line in BzK 4892 the natural width and fine-structure splitting are negligible compared to the velocity broadening of the Compton-thick reprocessor (FWHM $\sim 10^3\text{--}10^4 \text{ km s}^{-1}$, i.e. $\Delta\nu_{\text{src}}/\nu_{\text{src}} \sim 10^{-3}\text{--}10^{-2}$), and the line is in any case instrumentally unresolved at *Chandra*/ACIS spectral resolution. We therefore adopt the narrow-line idealisation $\Delta\nu_{\text{src}} \ll \nu_{\text{src}}$, under which the coherent core and the foam-broadened pedestal of Eq. (26) are spectrally separable; profile refinements (Voigt wings, Compton shoulder) would shift the threshold $\sigma_\phi^2 \lesssim 1$ by at most factors of order unity.

In the dynamical foam limit the suppression factor Eq. (24) reduces to $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$, which is the coherent fraction of the observed line profile via Eq. (26). Detecting the Iron K α feature requires $e^{-\sigma_\phi^2} \gtrsim e^{-1}$, i.e. $\sigma_\phi^2 \lesssim 1$. At this threshold the coherent line core carries fraction $e^{-1} \approx 37\%$ of the intrinsic flux — the minimum for a line detection not dominated by the broad pedestal, and intentionally conservative relative to the Maréchal criterion $\sigma_\phi^2 \lesssim 0.2$. The cumulative phase variance follows from the Ng–Perlman path-length model (1) via the global phase conversion $\sigma_\phi = 2\pi\sigma_\ell(D_C)/\lambda_{\text{obs}}$. The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ underlying Eq. (26) is satisfied here: at $\alpha = 0.59$ one has $\sigma_\ell \approx 1.2 \times 10^{-10} \text{ m}$ while the X-ray coherence length $c\tau_c = \lambda_{\text{obs}}^2/(2\pi\Delta\lambda) \approx 3 \times 10^{-8} \text{ m}$ for FWHM $\sim 10^3 \text{ km s}^{-1}$, giving $\sigma_\ell/(c\tau_c) \approx 4 \times 10^{-3}$; the envelope approximation is accurate to better than 10^{-4} . The phase variance, reproducing Ng & Perlman [4] Eq. (4), is

$$\sigma_\phi^2 \simeq (2\pi)^2 \frac{D_C^{2(1-\alpha)} \ell_P^{2\alpha}}{\lambda_{\text{obs}}^2}, \quad (33)$$

i.e., Eq. (23) with $A_\alpha = 1$ as in Ref. [4], where D_C is the comoving distance per Ng & Perlman Eq. (5). Imposing the threshold $\sigma_\phi^2 \lesssim 1$ and inverting,

$$\alpha \gtrsim \frac{\ln(2\pi D_C/\lambda_{\text{obs}})}{\ln(D_C/\ell_P)}, \quad (34)$$

the spectroscopic analogue of Eq. (8) of Ref. [4]. Inserting the numerical values $D_C \approx 1.88 \times 10^{26} \text{ m}$, $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10} \text{ m}$, and $\ell_P = 1.6 \times 10^{-35} \text{ m}$ gives $\ln(2\pi D_C/\lambda_{\text{obs}}) \approx 83.4$ and $\ln(D_C/\ell_P) \approx 140.6$, hence $\alpha \gtrsim 0.59$.

This bound, $\alpha \gtrsim 0.59$, is substantially tighter than the optical entry of Ref. [4] Table 1. Applying the threshold $\sigma_\phi^2 \lesssim 1$ (rms phase $\lesssim 1 \text{ rad}$) at $\lambda = 5000 \text{ \AA}$ and $D = 10 \text{ Gpc}$ gives $\alpha \gtrsim 0.55$, compared to their $\alpha > 0.53$. The improvement from 0.55 to 0.59 is not incidental: because $\sigma_\phi^2 \propto \lambda_{\text{obs}}^{-2}$, using the X-ray Fe K α line at $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10} \text{ m}$ instead of a typical optical line at $\sim 5 \times 10^{-7} \text{ m}$ increases the variance by a factor $\sim 5 \times 10^5$, making the threshold $\sigma_\phi^2 \lesssim 1$ far more constraining. The result does not rule out the holographic value $\alpha = 2/3$, consistent with all current optical and infrared imaging constraints. The bound $\alpha \gtrsim 0.59$ is weaker than the gamma-ray results of Section V A ($\alpha \gtrsim 0.65\text{--}0.66$) despite the much greater comoving distance ($D_C = 6.1 \text{ Gpc}$ vs. 0.63 Gpc). The gain for the gamma-ray probe is wavelength-driven: λ_{obs} for the 12.56 MeV line is ≈ 7000 times shorter than the Iron K α probe, boosting $\ln(2\pi D_C/\lambda_{\text{obs}})$ by ≈ 8.8 while the reduced distance decreases it by only ≈ 2.3 , for a net gain of ≈ 6.5 in the numerator of Eq. (34).

VI. CMB

VII. CONCLUSIONS

A single second-order coherence observable of spacetime foam emerges from this framework: the finite-lag Sievert discrepancy $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$, present when foam fluctuates on timescales shorter than the optical coherence time ($\tau_{\text{foam}} \lesssim \tau_L$). No foam-induced zero-lag bunching suppression arises in this model: at $\tau = 0$ the

same-direction jitter increments vanish identically and $g_{\text{obs}}^{(2)}(0) = 2$ for any extended source viewed by a single-mode detector (Section ??; Appendix A). Its value lies not in raw sensitivity — the signal is at $\sim 10^{-16}$ for the holographic $\alpha = 2/3$ model, well below current reach — but in three structural advantages over existing fringe-visibility and Strehl-ratio tests: (i) $\mathcal{S}(\tau)$ is constructed directly from the data, with $g_{\text{src}}^{(1)}(\tau)$ obtained from the source spectrum via Wiener–Khinchin, so a detection requires no separate source model; (ii) the angular-size suppression factor $\Phi_{\text{eff}}(r_{\perp}) \in [0, 1]$ (Section ??) makes extended targets weaker than compact ones but converts source-size variability across the target ensemble into a probe of the foam transverse coherence length $r_{\perp}$, a parameter inaccessible to single-target measurements; (iii) the framework remains sensitive to foam models with long-range transverse correlations that evade imaging tests entirely [10], and supplies a chromatic discriminant against plasma scattering through the $\sigma_{\phi} \propto 1/\lambda$ scaling.

The framework identifies high-priority targets: intrinsically compact emitters that saturate the upper bound regardless of $r_{\perp}$ (narrow-line Seyfert 1 cores, lensed-quasar BLRs), and high-flux transients (Type II_n supernovae, LBV giant eruptions) whose photon-rate advantage compensates for the modest cosmological lever arm at kpc–Mpc distances. A standing rapid-response HBT plan — covering η Car, any galactic core-collapse supernova, and any nearby ($D \lesssim 30$ Mpc) Type II_n identified by transient surveys — would let the framework exploit such high-yield events without prior allocation; transient targets with substantial angular extent contribute to the $r_{\perp}$ discriminant via Φ_{eff} . Strongly-lensed quasars (Section ??; J1330–0905, comparable to the NLS1 baseline after cosmological correction) round out the persistent target class.

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Appendix A: Pedestrian derivation of $g_{\text{obs}}^{(2)}(\tau)$

This appendix carries out the four-point function calculation underlying Section ?? step by step. The aim is to expose how the random variables (X, Y) that enter the connected Wick pairing arise, and to identify the regime in which the 2D average of Eq. (??) collapses to a 1D convolution (the point-source limit $\Phi = 1$) versus the regime in which the cross-direction carrier-phase factor $\mathcal{F}_{\text{cross}}$ of Eq. (??) appears.

1. Setup

The observed field is Eq. (??),

$$E_{\text{obs}}(t) = \int d\theta I^{1/2}(\theta) E_{\theta}(t - \delta t(t; \theta)), \quad (\text{A1})$$

with three statistical inputs:

1. Each E_{θ} is a stationary, zero-mean, complex-Gaussian (analytic-signal) process with unit mean intensity $\langle |E_{\theta}(s)|^2 \rangle = 1$.
2. Source spatial incoherence: $\langle E_{\theta}^*(s) E_{\theta'}(s') \rangle = g_{\text{src}}^{(1)}(s' - s) \delta(\theta - \theta')$, with anomalous pairings $\langle EE \rangle$ and $\langle E^* E^* \rangle$ vanishing.
3. δt is independent of E , and its joint distribution at distinct (t, θ) is the Gaussian process whose covariance is Eq. (4) divided by c^2 .

The brightness distribution is normalized, $\int I(\theta) d\theta = 1$, so $\langle I_{\text{obs}} \rangle = 1$ throughout this appendix.

2. Mean intensity

Plugging Eq. (A1) into $\langle |E_{\text{obs}}(t)|^2 \rangle$ and using spatial incoherence to collapse the angular integral to the diagonal,

$$\begin{aligned} \langle |E_{\text{obs}}(t)|^2 \rangle &= \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I^{1/2}(\boldsymbol{\theta}) I^{1/2}(\boldsymbol{\theta}') \langle E_{\boldsymbol{\theta}}^*(t - \delta t_1) E_{\boldsymbol{\theta}'}(t - \delta t_2) \rangle \\ &= \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) \langle |E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta}))|^2 \rangle = \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) = 1, \end{aligned} \quad (\text{A2})$$

where $\delta t_1 = \delta t(t; \boldsymbol{\theta})$ and $\delta t_2 = \delta t(t; \boldsymbol{\theta}')$. By stationarity, the random retarded time does not change the marginal second moment of each $E_{\boldsymbol{\theta}}$, so the foam preserves the mean intensity exactly: $\langle |E_{\text{obs}}|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$.

Appendix B: FIRAS Constraint on the Foam Exponent

1. Scope and Physical Motivation

This appendix does not present a completed FIRAS constraint on α ; rather, it identifies the physical inputs, derives the spectral template that any such analysis would fit, and outlines the analysis steps required to obtain a bound. The Wiener–Khinchin decomposition of Eq. (26) applies to any stationary field attenuated by the foam scattering kernel, not only to narrow emission lines. The Cosmic Microwave Background (CMB), observed across $\nu \in [6, 600]$ GHz by COBE/FIRAS, provides an independent broadband laboratory for this test: the same two-component signature — a suppressed coherent core and an incoherent pedestal (Section IV) — should appear in the CMB monopole spectrum if spacetime foam distorts photons on their path from the last-scattering surface. Because the foam signal is achromatic (geometric path-length jitter, not plasma dispersion), the frequency dependence of the foam-induced distortion follows $\sigma_\phi^2 \propto \nu^2$, distinct from standard chemical-potential (μ) and Compton- y spectral distortions. FIRAS constrains such distortions at the level $|\mu|, |y| \lesssim 10^{-5}$ to 10^{-4} .

2. Foam Phase Variance at CMB Frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_C \approx 14$ Gpc (Planck 2018 flat Λ CDM cosmology; $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). Equation (23) gives the accumulated phase variance directly at the comoving distance $D_C \approx 14$ Gpc:

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (\text{B1})$$

where ν is the observed frequency. The strong ν^2 dependence means foam effects are largest at the high-frequency end of the FIRAS band. Representative values at $\nu = 150$ GHz ($\lambda_{\text{CMB}} \approx 2$ mm):

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 6.8 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (\text{B2})$$

The strong ν^2 dependence of σ_ϕ^2 means foam effects are most pronounced at high frequency; for $\alpha = 1/2$ the variance reaches unity ($\sigma_\phi^2 = 1$) at $\nu \approx 574$ GHz, near the top of the FIRAS band, marking where the nonlinear regime becomes important. The achievable sensitivity threshold (accounting for degeneracies with μ and T_0) is $\alpha \lesssim 0.57$.

3. Derivation of the Observed CMB Spectrum

The narrow-band reduction of Eq. (26) — which underlies the spectral analysis of Section IV — was derived for quasi-monochromatic sources and does not apply to the broadband blackbody. Here we derive the foam signature for the CMB by starting from the full convolution integral and evaluating it in the fully dynamical limit $C_\alpha \rightarrow 0$.

Step 1: Saturated jitter variance

In the $C_\alpha \rightarrow 0$ limit, Eq. (10) gives

$$\sigma_\Delta^2(\tau) \rightarrow \frac{2\sigma_\ell^2(D)}{c^2} \quad \text{for all } \tau \neq 0. \quad (\text{B3})$$

The jitter variance saturates to a τ -independent value, twice the single-ray variance σ_ℓ^2/c^2 .

Step 2: Monochromatic decomposition

The blackbody field is broadband; we decompose it as an incoherent superposition of narrowband modes: $S_{\text{src}}(\nu') = B_{\nu'}(T_0)$ (spectral intensity per unit frequency). Each mode at frequency ν' is an independent stationary Gaussian process with coherence function

$$g_{\nu'}^{(1)}(\tau) = e^{-i2\pi\nu'\tau} \times [\text{envelope}(|\tau|)]. \quad (\text{B4})$$

Critically, each mode is independently phase-modulated by the *same* achromatic jitter $\delta t(t)$, which is frequency-independent by assumption (geometric path-length fluctuation, not plasma dispersion).

Step 3: Per-mode foam suppression

For a monochromatic mode at frequency ν' , the foam suppression factor is

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \text{where } \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D). \quad (\text{B5})$$

In the $C_\alpha \rightarrow 0$ limit, $\mathcal{F}_{\text{foam}}(\tau; \nu') \rightarrow e^{-\sigma_\phi^2(\nu')}$ for all $\tau \neq 0$ — a constant, τ -independent suppression.

Step 4: Two-component decomposition

Decompose the suppression factor into its asymptotic and residual parts:

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')} + [\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}]. \quad (\text{B6})$$

The first term is the $\tau \rightarrow \infty$ asymptote (the coherent core fraction); the second (the residual) vanishes at large $|\tau|$.

Step 5: Total first-order coherence

The intensity-weighted integral over modes:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{1}{I_{\text{tot}}} \int_0^\infty B_{\nu'}(T_0) e^{-i2\pi\nu'\tau} \mathcal{F}_{\text{foam}}(\tau; \nu') d\nu'. \quad (\text{B7})$$

Step 6: Wiener-Khinchin to the observed spectrum

Taking the Fourier transform via Eq. (25) and applying the two-component decomposition:

$$S_{\text{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu', \quad (\text{B8})$$

where $\hat{R}(\nu - \nu'; \nu')$ is the Fourier transform of the residual $\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}$.

Key observations:

- (i) The coherent core term $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$ is exact and model-independent; it requires only the assumptions of achromatic jitter and $C_\alpha \rightarrow 0$.
- (ii) The pedestal kernel $\hat{R}(\nu - \nu'; \nu')$ depends on the *short- τ behavior of $C_\alpha(\tau)$* — that is, on the foam coherence timescale τ_{foam} . Without specifying $C_\alpha(\tau)$, the pedestal shape is *undetermined*.
- (iii) The current placeholder kernel $K_\nu^{\text{fast}} = [\nu \ln(\nu_{\text{max}}/\nu_{\text{min}})]^{-1}$ does not correspond to any specific C_α model and should be replaced once one is chosen. The FIRAS bandpass [6, 600] GHz enters the analysis only as a likelihood window, not as the physical support of the redistribution kernel.

Photon-number conservation

The achromatic phase modulation $e^{i\omega\delta t(t)}$ at each frequency is unitary and time-reversal-invariant, so it conserves the photon number per mode: each photon is merely time-shifted, not absorbed or created. Therefore

$$\int_0^\infty \frac{S_{\text{obs}}(\nu) - B_\nu(T_0)}{h\nu} d\nu = 0 \quad (\text{photon-number conservation by achromatic unitarity}). \quad (\text{B9})$$

The two-component split $(e^{-\sigma_\phi^2} B_\nu) + \text{pedestal}$ satisfies this automatically if the pedestal is computed from the residual via Fourier transform.

Connection to Section IV.A

The derivation above is the broadband analogue of the assertion in Section IV.A that the complex amplitude decomposes as a $\delta(\nu)$ coherent-core term plus a broadened pedestal. A future revision should either lift this derivation into the main text or reference this appendix from Section IV.A for the broadband case.

4. Validity of the Fast-Foam Assumption

The fast-foam limit requires $\tau_{\text{foam}} \ll 1/\nu_{\text{CMB}} \sim 7$ ps. If the foam coherence time is set by the Planck scale, $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$ s, this condition is satisfied by forty orders of magnitude. For larger τ_{foam} , the foam suppression factor $e^{-\sigma_\phi^2[1-C_\alpha(\tau_{\text{foam}})]}$ (Eq. (24)) is reduced by the factor $[1-C_\alpha(\tau_{\text{foam}})]$, and the distortion amplitude is suppressed accordingly. In the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$, $C_\alpha \rightarrow 1$ and the CMB spectrum is unchanged, consistent with the vanishing of the HBT Siegert signal in that limit (Section III). In this limit the pedestal (residual term) also vanishes, consistent with Section B 3.

5. Degeneracy with Standard Spectral Distortions

The foam signal must be separated from two standard CMB spectral distortion templates. The chemical-potential (μ) and Compton- y distortions have characteristic shapes:

$$\Delta I_\nu^\mu = -\mu B_\nu(T_0) \frac{xe^x}{(e^x - 1)^2}, \quad (\text{B10})$$

$$\Delta I_\nu^y = y B_\nu(T_0) \frac{xe^x}{e^x - 1} \left[x \frac{e^x + 1}{e^x - 1} - 4 \right], \quad (\text{B11})$$

with $x = h\nu/(k_B T_0)$ and $T_0 = 2.7255$ K.

Comparing to the linearized foam template (high- ν deficit $\propto \nu^2 B_\nu$, low- ν pedestal undetermined), a key degeneracy emerges: the μ -distortion template's Rayleigh–Jeans limit is $\Delta I/B \propto -\mu/x \propto \nu^{-1}$, which is shape-degenerate with a $1/\nu$ -distributed pedestal. The current placeholder kernel $K_\nu^{\text{fast}} \propto 1/(\nu \ln[\nu_{\text{max}}/\nu_{\text{min}}])$ is precisely this form, maximizing the foam- μ correlation.

A Fisher matrix analysis (code: `fisher_firas.py`) using the linearized templates, diagonal FIRAS noise $\sigma_\nu = 0.01$ MJy/sr per channel, and 43 uniform channels from 60 to 600 GHz gives the correlation matrix shown in Table I and the marginal uncertainties

$$\sigma(A_{\text{eff}}) = 1.5 \times 10^{-17} \text{ m}^2 \quad (\text{marginal, all params free}), \quad \sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 2.6 \times 10^{-18} \text{ m}^2. \quad (\text{B12})$$

Marginalizing over μ and T_0 therefore degrades the foam amplitude uncertainty by a factor of ~ 6 , driven almost entirely by the foam- μ degeneracy.

TABLE I. Linearized Fisher matrix correlation coefficients ρ_{ij} for the FIRAS foam analysis with parameters $\theta = (A_{\text{eff}}, T_0, \mu, y)$. Computed using the placeholder pedestal kernel $K_\nu^{\text{fast}} \propto 1/\nu$; the dominant foam- μ degeneracy ($\rho = 0.976$) will change once a physical C_α model replaces this kernel.

	A_{eff}	T_0	μ	y
A_{eff}	1.000	0.937	0.976	0.010
T_0	0.937	1.000	0.969	-0.212
μ	0.976	0.969	1.000	-0.127
y	0.010	-0.212	-0.127	1.000

Table II gives the marginal detection SNR as a function of α (with $A_\alpha = 1$), showing that sensitivity falls below 1σ at $\alpha \approx 0.57$.

TABLE II. Marginal detection SNR for the foam amplitude A_{eff} as a function of foam exponent α (with $A_\alpha = 1$, $D_C = 13.87$ Gpc). “Marginal” marginalizes over T_0, μ, y ; “conditional” holds μ, y fixed.

α	$A_{\text{eff}} \text{ (m}^2\text{)}$	$\sigma_\phi^2(150 \text{ GHz})$	SNR (marginal)	SNR (μ, y fixed)
0.50	6.9×10^{-9}	6.8×10^{-2}	4.6×10^8	2.7×10^9
0.55	5.0×10^{-15}	4.9×10^{-8}	330	1920
0.56	2.9×10^{-16}	2.9×10^{-9}	19	113
0.57	1.7×10^{-17}	1.7×10^{-10}	1.1	6.7
0.58	1.0×10^{-18}	1.0×10^{-11}	0.07	0.40
0.59	6.1×10^{-20}	6.0×10^{-13}	0.004	0.024

These results use the placeholder kernel; the actual sensitivity limit depends on the physical pedestal shape. The foam- y correlation $\rho = 0.010$ is negligible: the y -distortion measurement provides essentially no help in separating the foam signal.

Action item: Repeat this Fisher analysis using (i) the FIRAS noise-plus-calibration covariance matrix from Fixsen (1996) in place of the diagonal approximation, and (ii) the full nonlinear foam template once $C_\alpha(\tau)$ is specified. Both changes will alter the correlations quantitatively, particularly the foam- μ degeneracy.

6. Bias in Recovered T_0 from Unmodeled Foam

The Fisher matrix provides two complementary measures of sensitivity. Section B5 asked *how precisely can we detect A_{eff} ?*; here we ask *how much would the fitted CMB temperature shift if foam were present but omitted from the model?*

Bias formula. Suppose the data contain a foam signal at amplitude $A_{\text{eff}}^{\text{true}}$ but the analyst fits only the standard parameters $\theta_{\text{fit}} = (T_0, \mu, y)$. In the linearized Fisher regime the resulting parameter shift is

$$\hat{\theta}_{\text{fit}} - \theta_{\text{fit}}^{\text{true}} = -(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}} A_{\text{eff}}^{\text{true}}, \quad (\text{B13})$$

where $F_{\text{fit,fit}}$ is the 3×3 lower-right block of the full Fisher matrix (for T_0, μ, y) and $F_{\text{fit,omit}}$ is the 3×1 cross-block coupling the fitted parameters to the omitted foam amplitude. Both blocks are extracted from the same matrix as Tables I–II, so no new computation is required.

Temperature bias. The T_0 component of Eq. (B13) is

$$\Delta T_0 \equiv \hat{T}_0 - T_0^{\text{true}} = b_{T_0} A_{\text{eff}}^{\text{true}}, \quad b_{T_0} \equiv -[(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}}]_{T_0}. \quad (\text{B14})$$

Numerically, $b_{T_0} = +1.21 \times 10^7 \text{ K m}^{-2}$: the bias is *positive*, meaning foam would cause the fitter to report a temperature slightly too high. This is counterintuitive but physically consistent. The placeholder pedestal $K_\nu^{\text{fast}} \propto \nu^{-1}$ produces a low-frequency flux excess that the fitter interprets as a temperature increase (since $\partial B_\nu / \partial T > 0$ in the Rayleigh–Jeans regime), while the high-frequency deficit is absorbed into a positive $\Delta\mu$. The dominant bias is in μ : numerically $b_\mu \approx 1.05 \times 10^8 \text{ m}^{-2}$, consistent with the strong foam- μ correlation $\rho = 0.976$ from Table I.

Bias-based bound. Requiring $|\Delta T_0| < \sigma(T_0) = 0.00057 \text{ K}$ (Fixsen 2009 [17]) gives

$$A_{\text{eff}}^{\text{max, bias}} = \frac{\sigma(T_0)}{|b_{T_0}|} = \frac{5.7 \times 10^{-4} \text{ K}}{1.21 \times 10^7 \text{ K m}^{-2}} = 4.7 \times 10^{-11} \text{ m}^2, \quad (\text{B15})$$

corresponding, with $A_\alpha = 1$, to a crossover exponent $\alpha_{\text{bias}} = 0.518$. Foam models with $\alpha < 0.518$ would have biased the Fixsen T_0 by more than 1σ had such a signal been present and unmodeled; models with $\alpha > 0.518$ do not.

Table III lists ΔT_0 and its ratio to $\sigma(T_0)$ for $\alpha = 0.50$ – 0.59 . At $\alpha = 0.57$ (where the marginal SNR ≈ 1 , Section B 5), the ratio $\Delta T_0/\sigma(T_0) \approx 4 \times 10^{-4}$: completely negligible. The published Fixsen $T_0 = 2.72548 \text{ K}$ is therefore not appreciably biased by any foam model near or above the FIRAS detection threshold.

TABLE III. Foam-induced bias in recovered T_0 when foam is present at amplitude A_{eff} (with $A_\alpha = 1$, $D_C = 13.87 \text{ Gpc}$) but omitted from the fit. Computed from the off-diagonal block of the Fisher matrix using the same noise and templates as Tables I–II; entries scale linearly with A_α . The column $\Delta T_0/\sigma(T_0)$ uses $\sigma(T_0) = 0.00057 \text{ K}$ (Fixsen 2009 [17]). The bold row at $\alpha = 0.518$ is the crossover where $|\Delta T_0| = \sigma(T_0)$.

α	$A_{\text{eff}} \text{ (m}^2\text{)}$	$\Delta T_0 \text{ (K)}$	$\Delta T_0/\sigma(T_0)$
0.500	6.9×10^{-9}	$+8.4 \times 10^{-2}$	147
0.518	4.7×10^{-11}	$+5.7 \times 10^{-4}$	1.0
0.530	1.4×10^{-12}	$+1.7 \times 10^{-5}$	0.030
0.550	5.0×10^{-15}	$+6.0 \times 10^{-8}$	1.1×10^{-4}
0.560	2.9×10^{-16}	$+3.6 \times 10^{-9}$	6.3×10^{-6}
0.570	1.7×10^{-17}	$+2.1 \times 10^{-10}$	3.7×10^{-7}
0.580	1.0×10^{-18}	$+1.2 \times 10^{-11}$	2.2×10^{-8}
0.590	6.1×10^{-20}	$+7.4 \times 10^{-13}$	1.3×10^{-9}

Comparison of bounds. The three α lower bounds from FIRAS are: (i) $\alpha \gtrsim 0.52$ (T_0 -bias bound, this subsection); (ii) $\alpha \gtrsim 0.57$ (marginal-SNR bound, Section B 5); (iii) $\alpha \gtrsim 0.59$ (BzK 4892 spectroscopic bound, Section V.A). The bias bound is the weakest because the strong foam– μ correlation ($\rho = 0.976$) allows most of the foam signal to be absorbed into μ without producing a large *individual* shift in T_0 . The same freedom that hides the bias inflates the marginal $\sigma(A_{\text{eff}})$, hence the SNR bound is tighter.

Caveat: The sign and magnitude of ΔT_0 depend on the placeholder pedestal kernel K_ν^{fast} . A physically derived pedestal from a specific $C_\alpha(\tau)$ model (Section B 3) may alter these values; repeating the bias calculation with the physical kernel is action item 2b of Section B 10.

7. Parameter Degeneracy Within the Foam Template

The foam template parameters α and A_α enter the linearized distortion only through the product $A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ — defining the combined amplitude A_{eff} . FIRAS constrains this amplitude directly, but not the exponent α separately. Converting a bound on A_{eff} to a bound on α requires fixing A_α by convention (e.g., $A_\alpha = 1$ following Ng and Perlman). Any reported foam constraint on α must state this assumption explicitly.

8. Nonlinear Regime at the Band Edge

The linearization $\Delta I_\nu \approx \sigma_\phi^2(\nu) \times [\text{template}]$ is valid only when $\sigma_\phi^2 \ll 1$. For $\alpha = 1/2$ (random-walk foam), $\sigma_\phi^2 = 1$ is reached at $\nu \approx 574 \text{ GHz}$, within the FIRAS band. At higher frequencies the linearization breaks down and the full nonlinear expression from Section B 3 must be used in the likelihood. The nonlinear curvature at high ν may help break the degeneracy with μ and y that dominates the marginal parameter space; this should be checked after computing the nonlinear Fisher matrix.

9. Relation to the BzK 4892 Bound

Section V.A reports $\alpha \gtrsim 0.59$ from the Fe K α spectral-Strehl test at $z = 2.578$. The FIRAS constraint achieves marginal sensitivity at $\alpha \lesssim 0.57$. These ranges do not overlap; FIRAS probes strictly lower α values (earlier in

cosmological time, longer propagation distance, higher cumulative variance at lower exponents). FIRAS therefore provides an independent systematics check using completely different inputs (absolute calibrated photometry, broadband rather than single-line, no source profile assumption) and probes the parameter space below the BzK bound where more exotic foam models would reside.

10. Proposed Analysis

The FIRAS constraint requires the following steps:

1. *Specify $C_\alpha(\tau)$ model.* Choose or derive the temporal decorrelation function (e.g., exponential with timescale τ_{foam} , power-law, or data-driven). Use Section B 3 to compute the pedestal kernel $\hat{R}$ from first principles.
2. *Compute nonlinear Fisher matrix.* Using the full nonlinear template (not the linearized approximation), compute the Fisher correlation matrix for $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_i)$ on the FIRAS noise-plus-calibration covariance. Assess how much the α - μ degeneracy is reduced once C_α is specified.
 - 2b. *Repeat bias calculation in the nonlinear regime.* Once the full nonlinear template is available, repeat the linearized bias computation of Section B 6 at the nonlinear level to verify that the sign and magnitude of ΔT_0 are reliable at the band edge ($\nu \gtrsim 400$ GHz, where $\sigma_\phi^2 \rightarrow 1$ for $\alpha \approx 0.50$ – 0.55). In the nonlinear regime the linearized formula Eq. (B13) must be replaced by a numerical comparison of maximum-likelihood estimators.
3. *Null-hypothesis validation.* Fit the FIRAS spectrum with $A_\alpha = 0$ and verify recovery of published values: $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL; Fixsen 1996, 2009).
4. *Joint Bayesian inference.* Sample the full posterior $p(\alpha, A_\alpha, T_0, \mu, y, \dots | \mathbf{I}^{\text{obs}})$ using the full nonlinear likelihood and FIRAS data. Report marginal bounds on α (fixing $A_\alpha = 1$ or marginalizing over it as specified).

11. Connection to the Spectral Framework of Section IV

Equation (??) in its original form asserts the two-component decomposition (coherent core plus pedestal) without derivation. The detailed calculation of Section B 3 is what grounds this decomposition: it shows how the broadband blackbody, when phase-modulated by achronatic jitter in the dynamical limit $C_\alpha \rightarrow 0$, naturally splits into a coherent-core term (exact, model-independent) and a pedestal (model-dependent on C_α). The FIRAS measurement probes this structure using the full 6–600 GHz dynamic range, complementing the single-line spectroscopy of Section V.A.

12. Likelihood and Inference Framework

The FIRAS monopole spectrum is compared to the full foam template (Section B 3) together with standard spectral-distortion templates (μ -distortion, Compton- y , CIB, and calibration modes) via a Gaussian log-likelihood:

$$\ln \mathcal{L} = -\frac{1}{2} \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r}, \quad (\text{B16})$$

where $\mathbf{r} = \mathbf{I}^{\text{obs}} - \mathbf{I}^{\text{model}}$ is the residual vector and $\mathbf{C}$ is the FIRAS noise-plus-calibration covariance matrix (Fixsen 1996). The free parameters are $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_1, c_2, c_3)$, where c_i are calibration mode amplitudes. A null-hypothesis validation fit with $A_\alpha = 0$ must recover the published FIRAS values $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL) before the foam parameter is freed for analysis.

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